

2023

COMPUTER SCIENCE — HONOURS

Paper : CC-12

(Object Oriented Programming)

Full Marks : 50

*The figures in the margin indicate full marks.**Candidates are required to give their answers in their own words
as far as practicable.*Answer **question no. 1** and **any four** questions from the rest.

1. Answer **any five** questions : 2×5
- (a) What is byte code?
 - (b) What do you mean by Java applet?
 - (c) Why is inheritance used in Java?
 - (d) What is protected access specifier?
 - (e) What is the role of garbage collection in Java?
 - (f) How does a vector differ from a list?
 - (g) Why is multithreading used in Java?
 - (h) What are the different types of AWT components?
2. (a) What are the different ways to create string object?
- (b) Explain the usage of `lastIndexOf (int ch, int fromIndex)` method in string operation.
- ~~(c)~~ Write a program that implements a runtime exception. — `ArrayIndexOutOfBoundsException`. 3+3+4
3. ~~(a)~~ What is the difference between Method Overloading and Method Overriding?
- (b) Explain Dynamic Method Dispatch with suitable examples.
- ~~(c)~~ Write a program in Java to access static variable and static method to explain 'static' keyword properly. 3+3+4
4. ~~(a)~~ When is a class in Java termed as abstract class? Give an example of an abstract class having an abstract method.
- ~~(b)~~ Write a program in Java to input a number and check whether it is Palindrome or not. 4+6

Please Turn Over

5. (a) How can thread prioritization be achieved in Java?
- ~~(b)~~ Differentiate between private and public access specifier with the help of an example.
- ~~(c)~~ State the differences between the string and stringbuffer classes in Java. 3+4+3
6. (a) What is the runtime exception?
- (b) What is the difference between *throw* and *throws*?
- (c) Design a user-defined exception handling program which will throw an exception if you entered a negative number. 2+2+6
7. (a) Differentiate between servlet and swing in Java.
- (b) Write a program in Java to find out the minimum element from an array without sorting it. 4+6
- ~~8.~~ (a) What problem can be faced due to multiple inheritance in JAVA? How can it be resolved?
- ~~(b)~~ What is a package? Explain the different access control packages in Java. 5+5
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